MIL-B-23459C(YD)
8 June 1981
SUPERSEDING
MIL-B-23459B(YD)
19 March 1976

MILITARY SPECIFICATION

BUOY, HOSE SUPPORT, INFLATABLE

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers an inflatable, rubber impregnated nylon fabric buoy.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116 - Preservation-Packaging, Methods of.
MIL-E-13072 - Enamel, Olive Drab, Lusterless, for Polychloroprene-Coated Fabric.
MIL-C-17415 - Cloth, Coated, and Webbing, Inflatable Boat and Miscellaneous Use.

STANDARDS

FEDERAL

FED STD No. 191 - Textile Test Methods.
FED STD No. 595 - Colors.
FED STD No. 601 - Rubber: Sampling and Testing.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 2050

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MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-794 - Parts and Equipment, Procedures for Packaging and Packing of.

DRAWING

NAVAL FACILITIES ENGINEERING COMMAND (NAVFAC)

889724 - Buoy, Hose Support, Collapsible, for Amphibious Fueling Facility.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting officer.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Description. The buoy shall be of single ply, rubber impregnated fabric construction with reinforcing hoops at each end, with steel end plate having an attached mooring ring and inflation-deflation port with cap, and with other features as shown on Drawing 889724 and specified herein. At the option of the contractor, and subject to approval by the contracting officer, design, construction, and fabric materials may vary from the details shown on the drawing, provided the finished product meets or exceeds the requirements for strength, water tightness, and resistance to wear described herein and on the applicable drawings.

3.2 First article. When specified (see 6.2), the contractor shall furnish a complete buoy for first article inspection and approval (see 4.2.1 and 6.4).

3.3 Design. The drawing, forming a part of this specification, is an engineering design drawing. The contractor is responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the item. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw

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materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.5 Construction. The buoy shall have the dimensions, characteristics, and construction details specified herein and as shown on the drawing. The fabric shall be single-ply nylon cloth impregnated with snythetic rubber compound, and shall conform to MIL-C-17415, type 9, class A, except that the coated cloth shall be uncured prior to vulcanizing of the buoy assembly. Seams shall be overlapped not less than 1 inch and the amount and method of lapping shall be such as to produce seams as strong as the coated cloth. Cement, gum strips, and chafing gum shall be of a vulcanizing type compatible with the rubber impregnated fabric. All seams, splices, and raw edges on the interior and exterior of the buoy shall be covered with gum strip not less than 3/4-inch wide. Foot flaps of the coated fabric shall be as shown on the drawing and shall be vulcanized to the outer side and bottom of the buoy. The assembled buoy shall be vulcanized as a unit in a steam or air pressure autoclave. After vulcanizing, the adhesion of seams and cemented areas shall be not less than 12 pounds (lb) per inch of width, and all requirements and characteristics of the coated fabric shall comply with MIL-C-17415. At the option of the contractor, the commercial type tire bead reinforcing hoops may be rectangular section having dimensions not less than 3/8 inch by 3/8 inch in lieu of 3/4-inch round bead shown on the drawing.

3.5.1 End plate assembly. At the option of the contractor, the end plate assembly parts shall be joined as shown on the applicable drawing, or shall be joined by welding provided that the end plate assembly is cadmium-plated after the welding operation. At the option of the manufacturer, the end plate assembly shall be installed in the end of the buoy either before or after vulcanizing of the buoy fabric. In either event, the method of joining the fabric to the metal shall develop a high strength bond and positive seal which will withstand the severe and repeated flexing and pulling experienced during normal use conditions in an ocean surf having waves up to 4 feet in height. This requirement shall be verified by the test described in 4.4.2. The flanged filler neck fitting and cap shall be nonvented type and the material shall be stainless steel. The cap gasket shall be 1/8-inch neoprene rubber. The end plate assembly shall be cadmium-plated before assembly to the filler neck.

3.5.2 Repairs. Damage occurring before curing shall be repaired before curing. The hot-patch method shall be used to repair damages after curing. Patches shall have an adhesion not less than that specified in 3.5 for seams and cemented areas. Not more than three patches will be permitted in the buoy body. Repairs shall restore the full strength of the coated fabric, or the seam, as applicable.

3.6 Performance.

3.6.1 Leakage. The buoy shall not leak when fully inflated and the cap installed, as demonstrated by the test specified in 4.4.4.

3.6.2 Storage temperature range. The buoy shall be suitable for storage in a temperature range from -40° Fahrenheit (F) to +160°F. The buoy body material shall not flake or crack in any ambient temperature to -40°F, and shall develop no damage or tackiness in any ambient temperature to +160°F.

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3.7 Vulcanizing repair items. When specified (see 6.2), the vulcanizing repair items shown in Table I shall be furnished.

TABLE I. Vulcanizing repair items.

Item	Description	Quantity
Patching, 12 inches by 12 inches	Cured coated fabric conforming to 3.5	1 piece
Sealing strip, 1 inch by 60 inches	Cured gum strip conforming to 3.5	1 strip
Cement	Room temperature curing, with accelerator as required for rapid curing	Two 1/2-pint cans

3.8 Workmanship.

3.8.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.8.2 Riveted connections. All rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall completely fill the hole. Rivet heads shall be of an approved shape and shall be concentric with the rivet. Rivet heads shall make full contact with the intended surface. The rivet shall be so fabricated as to develop a joint strength not less than the design value.

3.8.3 Welding. Surfaces to be welded shall be free from foreign matter which would be injurious to the weld. Welding procedures shall be in accordance with a nationally recognized code. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit imposed stresses without permanent deformation or failure when subject to proof or service loadings.

3.9 Cleaning. The interior and exterior surfaces of the buoy shall be free of dirt, talc, separation sheet, excess material used in construction of buoy, and other foreign material.

3.10 Color. Color of the exterior surfaces of the buoy shall be black. The color may be integral with the coating compound, or may be obtained by coating with enamel conforming to MIL-E-13072, except the color shall be Black, No. 37038 of FED STD No. 595. Dry film thickness of the paint coating shall be not less than 1 mil.

3.11 Identification marking. The buoy shall be marked for identification by stenciling in a visible location with the following information:

BUOY, HOSE SUPPORT, INFLATABLE
MIL-B-23459C(YD)
(DATE OF MANUFACTURE)

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The marking shall be as permanent as the normal life expectancy of the buoy and shall be capable of withstanding the environmental tests specified for the buoy.

3.12 Servicing and restoration. Each unit tested shall be serviced and restored to a service condition equal to the original condition of the unit, neglecting nominal wear incurred during the tests. The restoration shall include paint touchup or repainting, as required for delivery.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2.1).
(b) Quality conformance inspection (see 4.2.2).
(c) Packaging inspection (see 4.5).

4.2.1 First article inspection. First article inspection shall be performed on one buoy. This inspection shall include the examination of 4.3 and all the tests of 4.4. The first article shall be the first buoy produced prior to initiation of the production items.

4.2.2 Quality conformance inspection. The quality conformance inspection shall consist of the examination of 4.3, the tests of 4.4.3 and 4.4.4, and the inspection of 4.5.

4.3 Examination. Each buoy shall be examined for compliance with the requirements specified in Section 3 of this specification. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests.

4.4.1 Temperature tests. Resistance of the coated fabric to high and low temperatures shall be determined in accordance with FED STD No. 191, methods 5872 and 5874 respectively, with the following exceptions: (1) the high temperature shall be 160°F ±2°F; (2) the low temperature shall be -40°F ±2°F; and (3) the specimens for the tests shall be conditioned in the required environment for a period of 24 hours. The coated fabric shall show a rating of not more than 2 when tested by method 5872.

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4.4.2 Dynamic buoyancy test. The test arrangement shall be similar to that shown on Figure 1. The mechanism shall be powered and rigged to alternately submerge and float the normally inflated buoy in the tank of water at a rate of 8 to 10 complete cycles per minute for not less than 48 hours. At maximum submergence, the top of the buoy shall be not less than 3 feet below the water surface. Upon examination after this test, if the buoy shows indications of seam or joint failures or loss of buoyancy of more than 5 percent, the buoy shall be rejected.

4.4.3 Seam strength test. Two seam specimens, 1-inch wide, shall be prepared from each day's production and cured simultaneously with a buoy or group of buoys. The adhesion of seams and cemented areas of the specimens shall be tested in accordance with FED STD No. 601, method 8311, using a rate of travel for the power grip of 2 inches per minute.

4.4.4 Leakage. Each buoy shall be manually inflated by extending the body section to the full height with the cap removed, and then replacing the cap while holding the buoy in its extended position. The buoy shall then be totally submerged in a suitable container, using clear water as the liquid medium. While submerged, the buoy shall be compressed using a static load or a manual pressure of 100 lb. Leaks detected in any portion of the buoy shall be repaired and the buoy retested. At the option of the contractor, an internal air pressure of 1/2 lb per square inch gage shall be applied to the buoy and a soapsuds method of testing for leaks may be employed, provided the 100 lb static load is applied and all external surfaces of the buoy are coated with soapsuds and carefully examined for leaks.

4.5 Packaging inspection. The preservation, packing and marking of the buoys shall be inspected to verify conformance to the requirements of Section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Buoys. Each buoy shall be deflated and packaged in accordance with MIL-P-116, method III.

5.1.1.2 Repair kits. Each repair kit, when furnished, shall be packed in accordance with MIL-P-116, method III, in a fiberboard box conforming to PPP-B-636, weather-resistant.

5.1.2 Commercial. The buoys and repair kits, when furnished, shall be packaged in accordance with the contractor's standard practice.

5.2 Packing. The packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. The buoys and repair kits, when furnished, shall be packed in accordance to the level A requirements of MIL-STD-794 utilizing applicable containers selected from table II.

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5.2.2 Level B. The buoys and repair kits, when furnished, shall be packed in accordance to the level B requirements of MIL-STD-794.

5.2.3 Commercial. The buoys and repair kits, when furnished, shall be packed in a manner which will insure arrival at destination in satisfactory condition. Containers and packing shall conform to the applicable carrier rules and regulations.

5.3 Marking. Interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The inflatable buoy is used to support fuel lines of amphibious fueling facilities.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) When first article is required for inspection and approval (see 3.2, 4.2.1, and 6.4).
(c) When vulcanizing repair items are required (see 3.7).
(d) Level of preservation and level of packing required (see 5.1 and 5.2).

6.3 Contract data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the DAR. The first article should be preproduction sample. The first article should consist of one unit. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examination, test, and approval of the first article.

Preparing activity:

Navy - YD

Project No. 2050-N025

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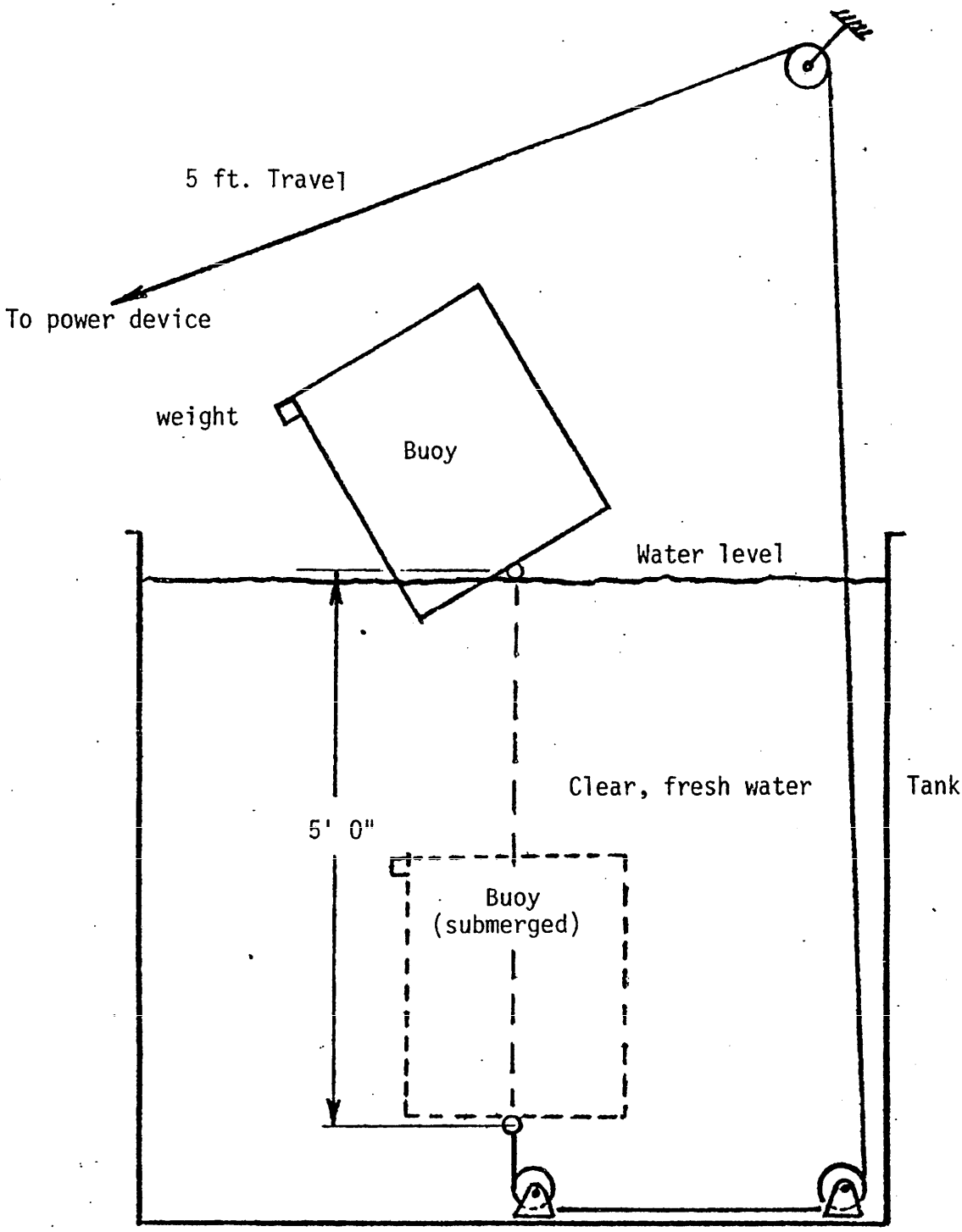


Figure 1. Dynamic buoyancy test.

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2050-N025

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-B-23459C(YD) BUOY, HOSE SUPPORT, INFLATABLE

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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